

The BT operational support systems architecture

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This paper outlines the process and systems framework that BT has developed to assist in the design, development and procurement of operational support systems.

1. Introduction

BT currently invests hundreds of millions of pounds per annum in the development, procurement and maintenance of its operational support systems (OSS). In order to ensure good value for that investment, the design community needs an OSS design framework or architecture which will help in the production of good designs. The telecommunications management network (TMN) is the architecture currently accepted by the industry for this purpose [1]. However, TMN has focused primarily on interworking between operators and between systems and networks. It does not currently provide adequate help in the implementation of system solutions for network and service management internal to an operator.

In order to overcome this, BT has developed an improved process and systems architecture to assist in the design, development and procurement of operational support systems solutions for the management of all of its services and networks. This architecture (see Fig 1) is comprehensive in considering all processes, functions and data within network and service management, and can be considered to be complementary to TMN.

2. The time for change

The TMN architecture, derived primarily from BT's cooperative networking architecture (CNA [2]), was designed in the 1980s when telecommunications was very much simpler than it is today. The telecommunications environment is becoming increasingly more complex and demanding on network and service management with:

- rapid expansion of services,
- faster introduction of new services,
- increasing network complexity:
 - overlay networks,
 - multiple technologies,

- regulation,
- competition,
- globalisation.

The problem is that almost every facet of this complexity impacts on the operator's OSS. In order to deal effectively with this dynamic and complex environment, the design and development environment and the OSS themselves need to be well structured and highly flexible, providing solutions in shorter time-scales at low cost. Often, a small-scale solution is required to suit the introduction of new niche services which may subsequently grow to match the market demand for the service and become integrated into the main operational processes of the operator.

TMN has concentrated on looking outwards covering, primarily, interworking between operators and interworking between OSS and the network. Little has been done to address the design of OSS solutions required for the core of network and service management within an operator's or service provider's company. It provides little detail of the functions and data at each of the layers and almost no model of the relationships between the functions and data. It also gives no view of the processes, although this area is now being addressed by the SMART initiative within the NM forum [3].

3. The legacy

This lack of a complete, coherent design framework has resulted in poor solution designs, which, over the years, has created a legacy in BT of:

- over 1000 separate systems,
- islands of automation with little integration or flow-through of processes,
- stove-pipe solutions for particular services or networks with high levels of duplication,

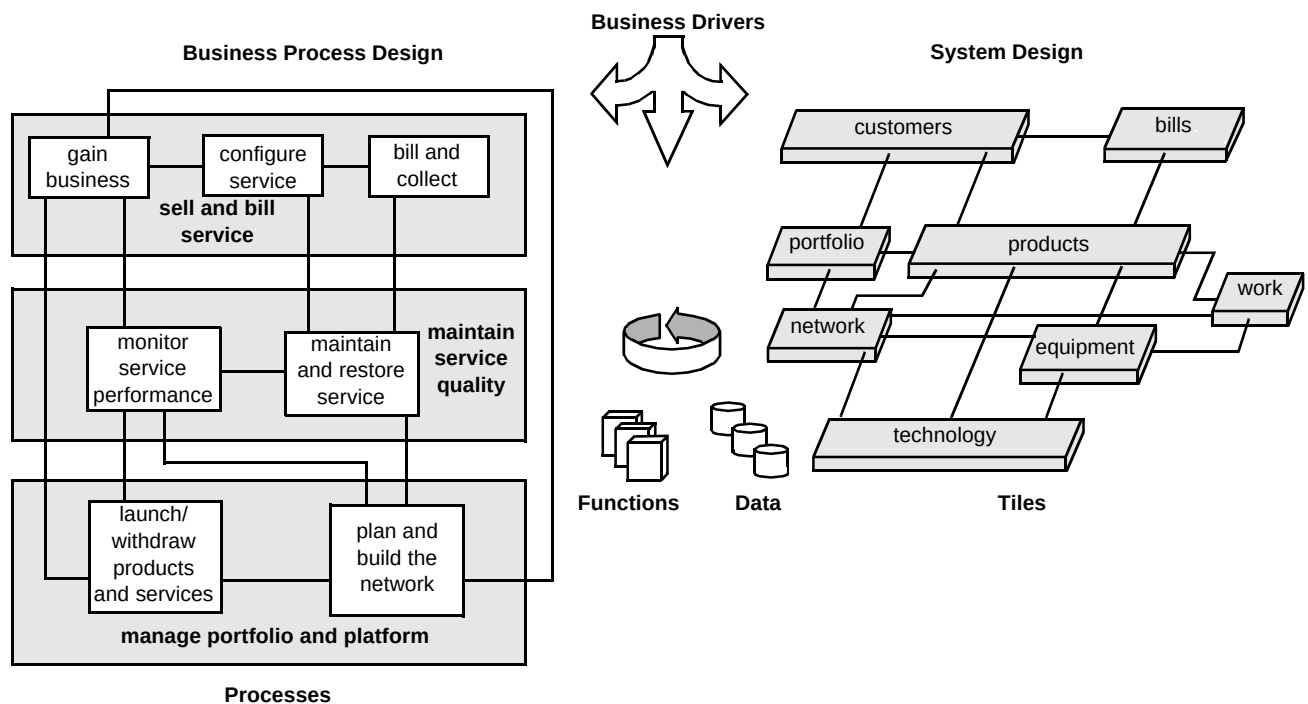


Fig 1 The BT OSS architecture.

- poor data quality due to duplication and limited access to the data,
- multiple computing platforms increasing the problems of integration,
- high cost of ownership in delivery and ongoing running costs.
- maintain service quality — responsible for ensuring that the quality of each instance of service sold is within the service level agreement with the customer,
- manage portfolio and platform — responsible for managing the portfolio of products and services which the company sells and the network and OSS platforms on which those services are delivered.

4. Process driven

In order to ensure that systems deliver the functionality that is required by the company, BT is committed to an approach where solutions should be driven from a process perspective with operational requirements being expressed in process terms. This architecture therefore includes a set of fully defined business processes with all significant interactions between the business processes. The objective is to use this process architecture for the design and evolution of specific operational process such that, with time, those operational processes become common and standardised across the business. In addition to facilitating common OSS solutions, it should also simplify the problems of operational organisation design and the integration of the management of new services and technologies into existing organisation structures. The other aspects of the architecture have been derived from this process model (see Fig 2).

The process architecture is comprised of three very high-level processes which have been called key business processes

- sell and bill service — responsible for proactively selling services to customers and collecting the revenue for the use of those services,

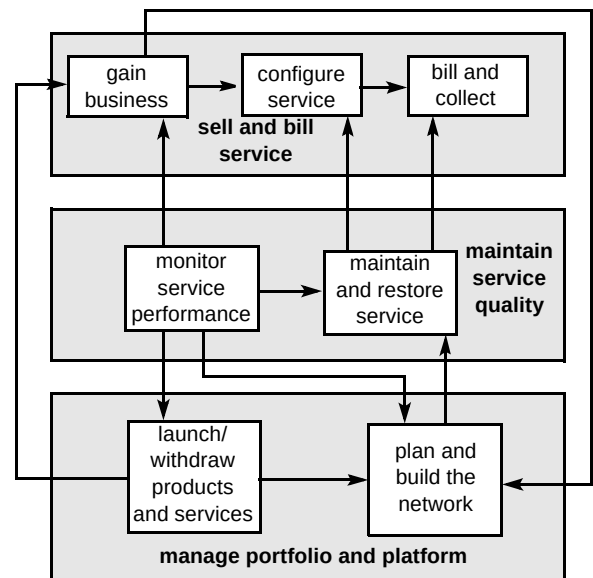


Fig 2 The process architecture.

Within each of these key processes are a number of lower level business processes, seven in total, each of which is called a generic business process (GBP), and within each of these are a set of interlinked component processes. These component processes themselves have been decomposed

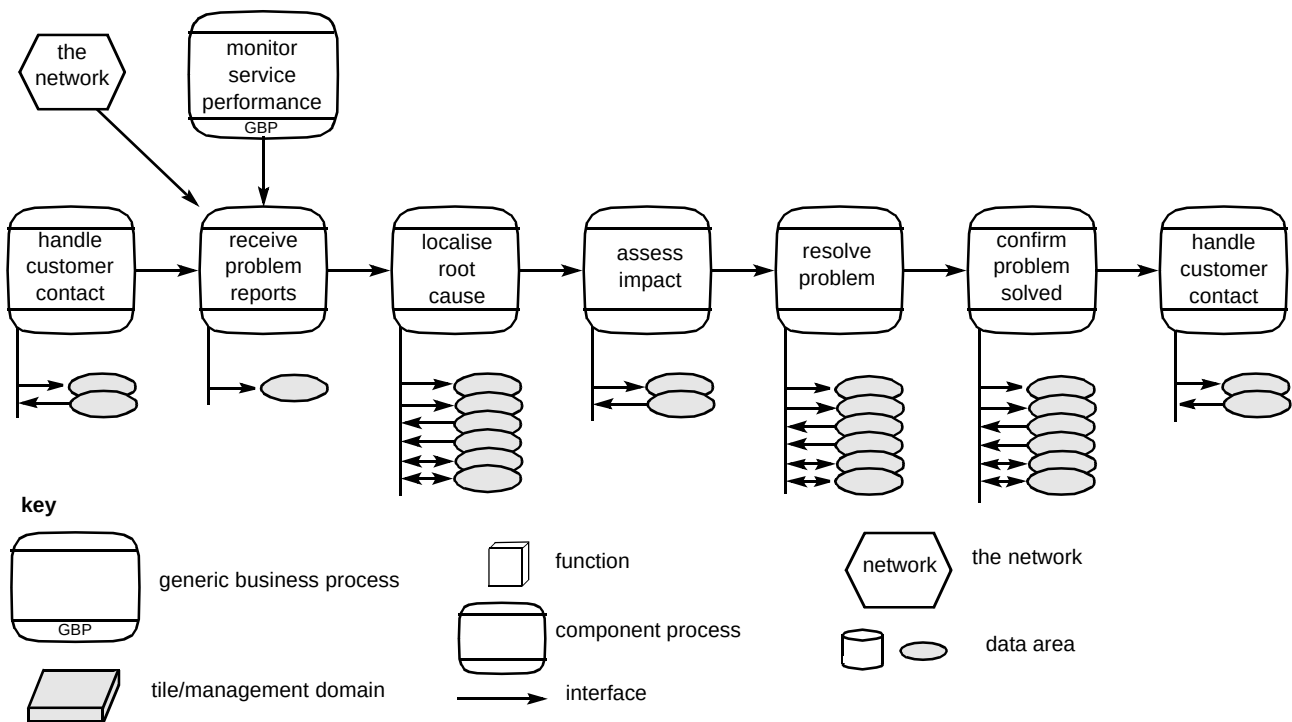


Fig 3 Maintain the network generic business process.

one further level into what have been termed actions. Most importantly, the generic business processes also contain all of those items of data on which the component processes operate. This is considered to be a most vital aspect of the process architecture.

As an example, a simplified view of one of these generic business processes is shown in Fig 3, including the more significant component processes, their interfaces and their interaction with the various items of data; Fig 3 also includes a key to the symbols used throughout this paper.

The component processes and actions were further analysed and a set of functions were defined which will deliver the requirements of the processes. The objective was to identify a set of reusable components of a scale and significance which could be considered to be the future building blocks of systems. The functions will be delivered by applications software in the operational support systems. common functions will deliver parts of several different business processes.

Each component process within the business process operates on specific items of data and many of the component processes operate on the same data items. More importantly, component processes in different business processes operate on the same data items. This complete view of these relationships is vital in understanding the issues of dependencies between the systems which deliver these capabilities and the importance of making the data accessible. Historically, systems have been designed which implement specific processes or parts of processes together with the required data without making that data accessible

to other systems. This has led to duplication of the data leading to inconsistency.

5. Data is key

Data is now considered to be the single most important aspect of OSS design, but in the past this has been given insufficient attention. In order to address this problem, BT has analysed all of the data required within service and network management and created a structure within which to manage it. From analysis of the processes, approximately 35 types of data (called data areas) were identified, defined and grouped into six data groups, shown in Fig 4.

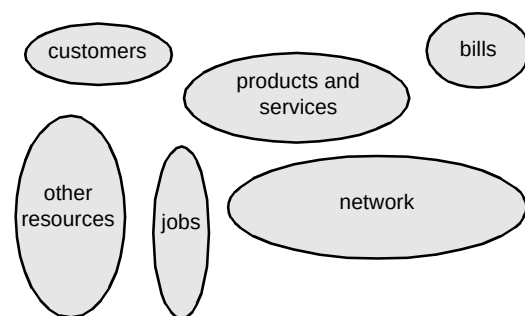


Fig 4 The six data groups.

- The customers group contains data about customers including name and address, role in a company, socio-economic group, what services they have, etc, together with a record of all of BT's interactions with them, including current orders, problems and billing queries.

- The bills group includes customer accounts, invoices raised for use of services, information about billing problems, billing profiles and billing schedules which determine when invoices should be raised.
- The jobs group contains information about key activities which are managed by the job control functions, including orders, trouble tickets and build orders. Some of the information in jobs will be manual tasks.
- Other resources contains information about work, the schedule of manual tasks, the people, including skills and whereabouts and security information about people's access rights to systems and buildings including BT people and customers.
- The products and services group of data describes the products and services, both in terms of the portfolio of types of service sold and the instances of services which have been sold to specific customers. In more detail below (see Fig 5) the products and services portfolio contains information about each product and service in the portfolio including its service features and charging structure. Associated with this are marketing plans for promotion of specific products, market intelligence collected from customers about products and requirements and product plans for the launch of new products.

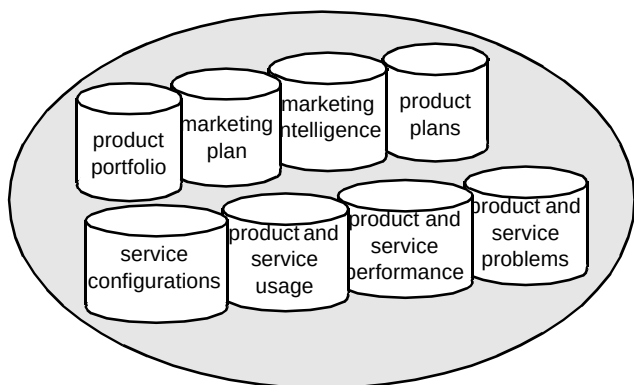


Fig 5 The products and services data areas.

Service configurations contains high-level information about how the features of each instance of service are delivered by the network together with the service level agreements (SLAs). Associated with these are behaviour data about types and instances of service including usage, performance and problems.

- The network group (see Fig 6) contains data which describes the structure and behaviour of the network. The structure of the network is described in terms of topologies, configuration and inventories. The behaviour data includes information about usage, problems and performance and are associated with the relevant structure data. Finally, there are the planning policy and rules.

This structuring and grouping of the data will help in the design of databases and data servers due to the understanding of the relationships and affinities between the data areas and with the processes or functions which operate on them.

6. Physical systems approach

The above description demonstrates the approach taken in designing the logical components of the process and data architectures. This logical architecture has been designed to ensure that all future OSS solutions deliver known and required functionality and have appropriate access to a unique set of properly managed data. This was used as the starting point for the development of a physical systems architecture.

This physical architecture has been designed on the following basis.

- Object-based (data-centred) approach — whereas previous architectures have been primarily functionally (or process) based, this architecture puts a much higher emphasis on data and avoiding the data problems in the legacy systems.
- Identify the key things to be managed — the best set of high-level business objects which the business needs to manage have been identified, considering where changes are likely to occur to avoid too many different types of change impacting on the same systems.
- Identify management domains — knowing the objects, management domains have been scoped and characterised within which the objects will be managed.
- Identify the data which describes the objects — from the data analysis work, the data has been identified which describes those objects.
- Identify the functions which operate on the data — from the process analysis work, the component processes and the functions which operate on that data were identified.
- Encapsulate the functions and data within domains — the data and its functions were grouped together or encapsulated within the domain.
- Minimise sharing of data across domains — with the functions and the data they need held together within the domains, there is little need for sharing of data across domains.
- Maximise reuse of concepts, designs, code, interfaces — within the domains, there will be opportunities to reuse concepts in terms of structures and relationships, designs, applications, database schemas and interfaces.

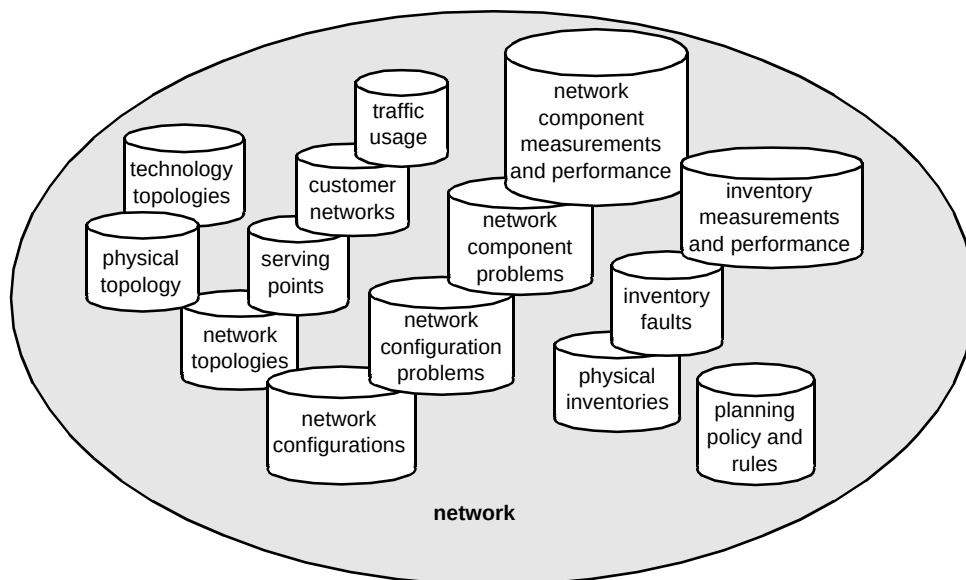


Fig 6 The network data areas.

7. OSS management domains

Using the above approach, a set of eight management domains were developed, based around the previously defined data groupings. These management domains are shown in Fig 7.

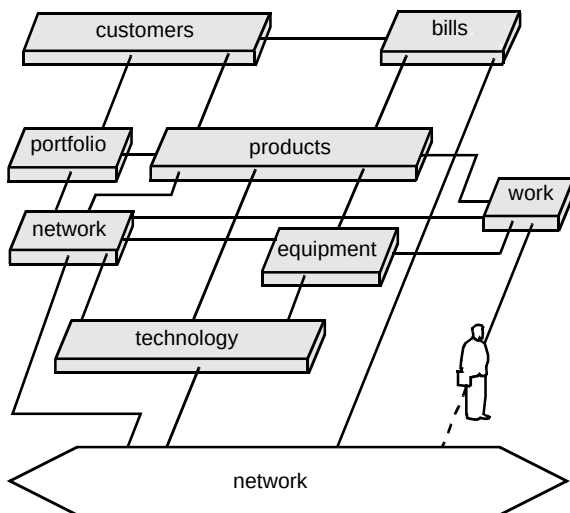


Fig 7 The OSS management domains (tiles).

- The customer domain deals with all aspects of BT's customers and contains all of the information about customers and their interactions with BT. It also contains the functions which support communication with the customer.

In the data architecture, the products and services group contained both types and instances of services. Here they are separated as the issues in managing them are quite different:

- The product domain is about the instances of all products that have been delivered to customers, the SLA for each of them, how they were realised and how they have performed. It also contains the functions for delivering the services, managing problems and analysing performance and usage.
- The portfolio domain is about the range of products and services which BT sells, their performance, market intelligence and plans for new products. It also contains the functions for designing, performing trials for, and launching new products.

The network data in the data architecture has been divided into three separate domains (technology, network, equipment).

- The technology domain is about the individual technologies which make up the complete network platform and how the capabilities of the technologies have been configured to deliver instances of services. It contains the functions to assign capacity, configure the technology, plan capacity changes, analyse problems and test the network.
- The network domain is about the combination of various network technologies into a service-bearing platform and covers network design, analysis and resolution of problems and management of traffic. The network and technology domains deal only with the logical functions, features or capabilities of the network.
- The equipment domain is about the physical equipment which provides the functions to support the products and services, planned changes, faults and performance of the physical equipment.

- The work domain is about the manual work which needs to be carried out on the network and the workforce which performs that work. It contains functions to schedule and allocate work to the workforce.
- The bills domain is about the customer accounts, invoices for usage of products, and billing problems. It contains the functions to raise charges for usage, generate invoices, and handle payments, problems and debts.

These eight domains contain all of the functions and data to support the requirements of the seven generic business processes. The functions and data will be realised by a set of applications and data servers within each domain. These applications and data servers will be supported by a set of closely coupled computers forming a computing infrastructure.

The domains are now known as tiles due to the icon that has been used to depict them.

8. OSS tiles — segmentation

Some of the tiles have been segmented, supporting different physical solutions where there is a valid business requirement. The logic of the tile will remain the same in each case. For example, a complete network will consist of many different managed technologies which will require different management solutions. Parts of the relevant technology tile segment will be supplied by the technology supplier.

The customer tile is also segmented as the physical solution will be different for different customer segments (see Fig 8).

9. Tile detail

Each of the tiles contains a set of tightly coupled functions and data. In the case of the product tile (see Fig 9), there are ten data stores: products and services portfolio, product and service type performance, type usage and type problems, product and service instance measures and performance, instance usage and instance problems, service configurations, orders and serving points. These data stores are surrounded by and support nine functions which handle the collection, analysis and reporting of product type performance and usage. Order taking, service solution design, job control and product test management are involved in the 'gain business' and 'configure service' processes. Product test management is also involved in the 'maintain and restore' service process together with service impact assessment and product problem analysis.

There is a high level of interconnectivity between these functions and data stores, reflecting the strong process and data affinity, and supporting the existence of the tile.

There are also a number of linkages to functions on other tiles.

The other tiles (for example, the technology tile — see Fig 10) have a similar set of functions, data stores and interfaces. The functions and data on the tile and across tiles co-operate to deliver the end-to-end requirements of the business processes.

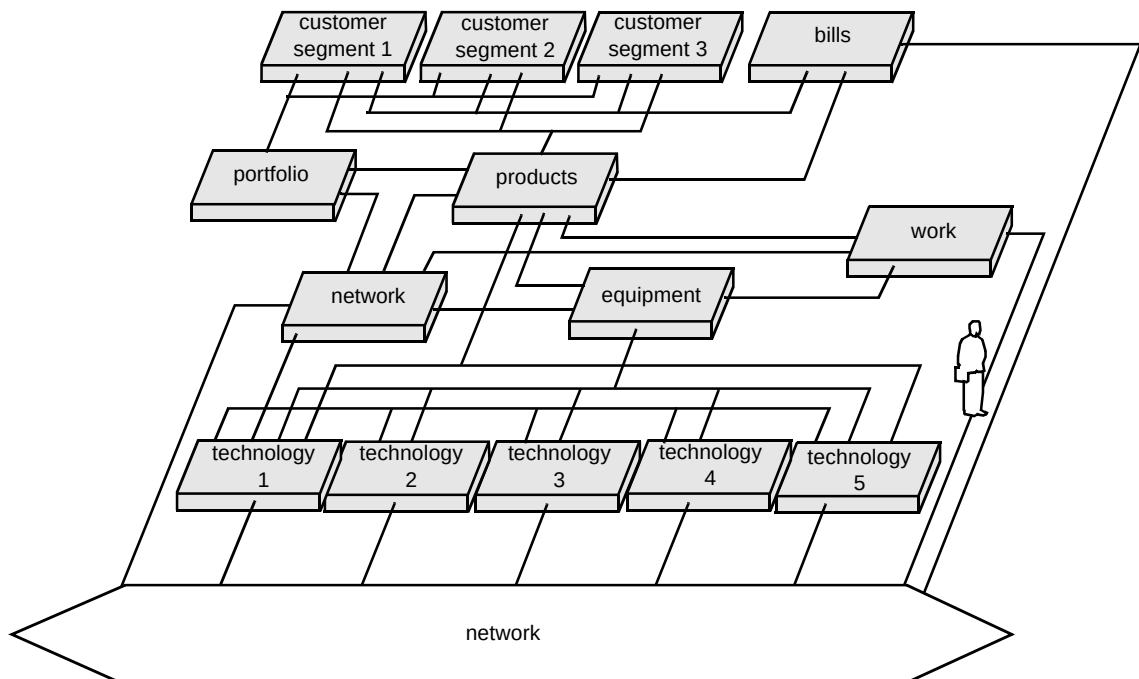


Fig 8 Tiles segmentation.

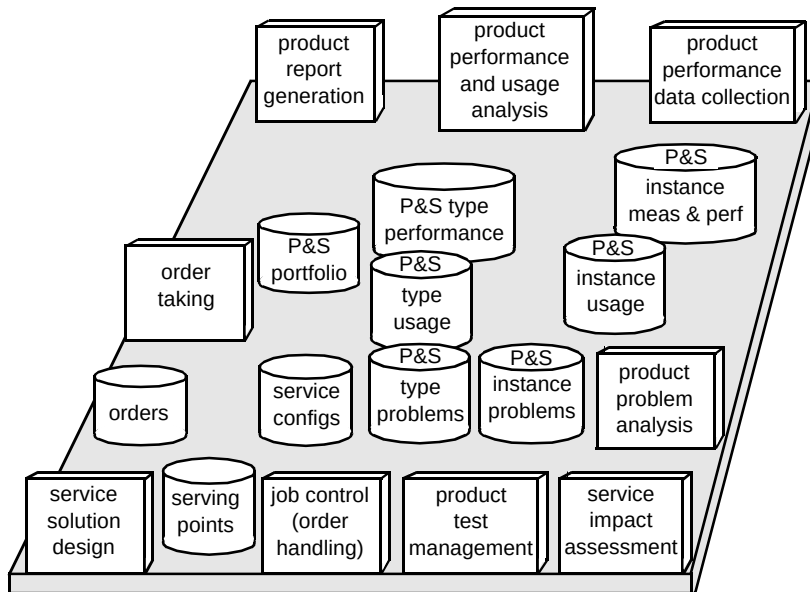


Fig 9 The product tile.

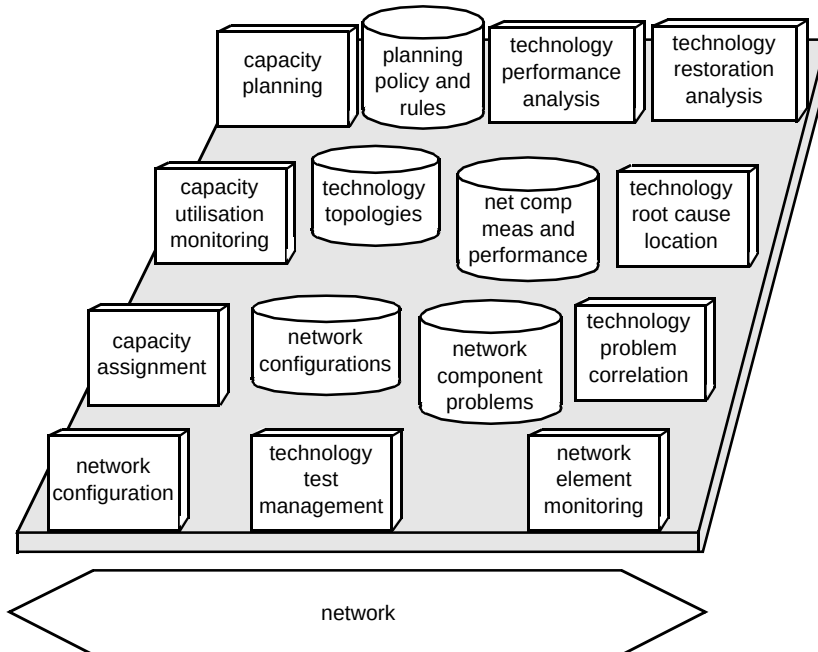


Fig 10 The technology tile.

10. Mapping to TMN

Clearly, much of the rest of the industry in terms of operators and suppliers are committed to some degree to TMN. BT still supports the TMN work, but would like to see improvements made to address the deficiencies in TMN, which do not help in the design of OSS solutions for the current complex telecommunications environment, but which have been addressed by this new architecture.

Work on this has been started by understanding the relationship between them, as shown in Fig 11.

11. Using the architecture

There are three main ways in which BT will use the architecture shown in Fig 12.

For each of BT's current or legacy systems, the system can be characterised in terms of the functions it performs and the data which it holds or uses. This will enable that system to be placed on the appropriate tile (or tiles) in the architecture. Legacy systems should not necessarily be expected to align with the architecture and it will often be found that a legacy system has an appearance on more than one tile. In future, with new systems, the aim is to avoid this.

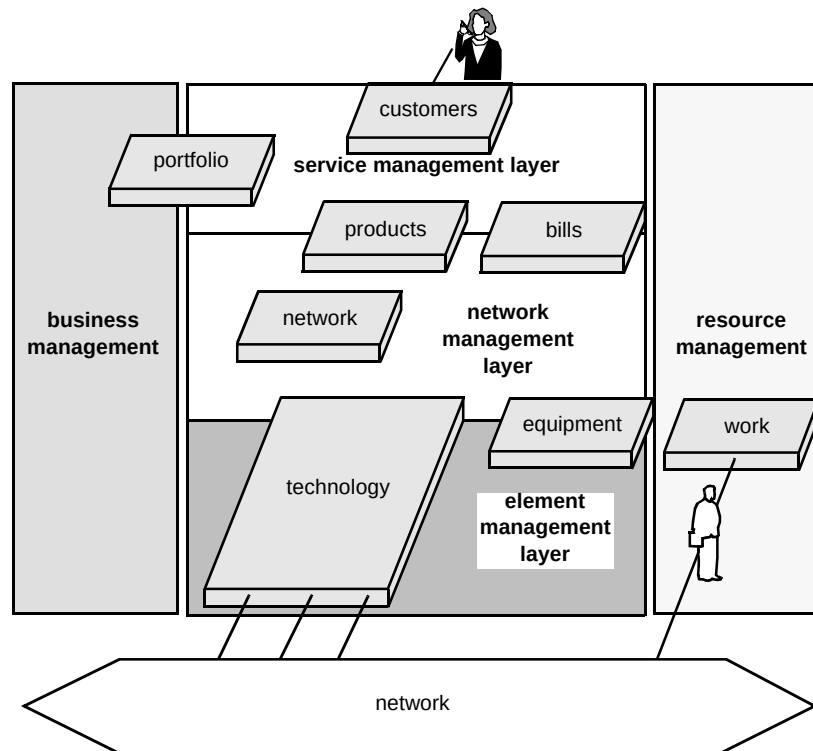


Fig 11 Relationship with TMN.

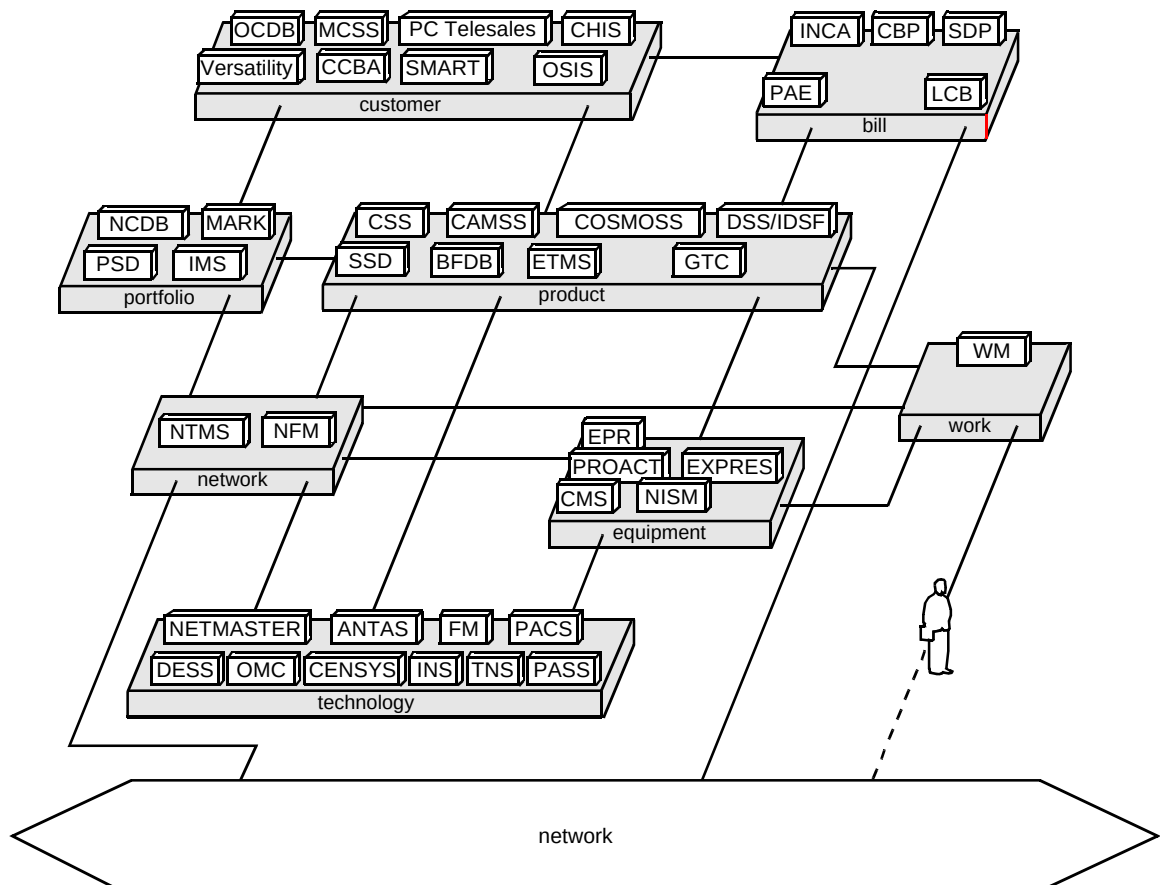


Fig 12 BT's key systems by tiles.

When all current systems have been placed, decisions can then be made about their evolution and, in many cases, their rationalisation. An evolution plan has been produced for each of the tiles showing how the current systems will evolve and how gaps in functionality and data will be resolved.

The third and probably most important use will be in the design of solutions and systems for new business objectives and initiatives. The architecture will help to ensure that each solution is consistent with others and that there is maximum reuse of solutions. It is a holistic architecture, covering the whole of the service and network management scope, enabling designs to be produced for specific problems with a knowledge of the broader context. It also provides an essential checklist of all of the issues which need to be considered in a design.

Figure 12 shows BT's key systems mapped to the relevant tiles, having characterised them by function and data types supported.

This architecture will help BT to move its whole systems base forward in a controlled way by designing coherent solutions from a consistent understanding of the complete OSS picture used by all of its designers.

12. Conclusions

This paper has given an overview of BT's OSS architecture, explaining why it was developed, giving a brief description of its contents, and illustrating how it fits in with other initiatives. It is envisaged that use of the architecture will lead to a reduced number of operational systems and more controlled data, thereby allowing greater management control of BT's investment in IT. In effect, it will help BT to move its whole systems base forward in a controlled way by designing coherent solutions from a consistent understanding of the complete OSS picture used by all of its designers.

Pyzer et al [4] go into more detail about the wider framework into which this architecture fits, and give more insight into how BT will use it to assist in the design, development, procurement and deployment of computer systems.

References

- 1 Network Management Forum: 'TMN Basic Management Platform', NMF CS302, (1994).
- 2 'Co-operative Networking Architecture: CNA Technical Guide: Introduction to Network Management', BT (1990).
- 3 Network Management Forum: 'SMART: OMNIPOINT2', (1995).
- 4 Pyzer S, Williams P and Richards D: 'Applying the BT operational support systems architecture', BT Technol J, 15, No 1, pp 22—28 (January 1997).



Nick Furley graduated with a BSc in electrical and electronic engineering from the University of Manchester in 1974 within the BT student sponsorship scheme. He worked on the design and delivery of a number of BT's operational support systems including the digital exchange support system (DESS) and the district data collector (DDC), now known as the network mediation processor (NMP). In 1988, he took responsibility for the architecture and evolution planning of the network management systems and set up the network control architecture board. In 1994, he was seconded to the Breakout project where he was a member of the team which developed the OSS architecture framework described here. In 1966, he took on the broader scope of architecture and evolution planning for all OSS.